

PALLAB DEY

Sylhet, Bangladesh (Open to relocation) | +880-1751164660 | pallab@pallabdey.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Computer Science graduate (CGPA 3.96/4.00) turning complex datasets into decisions companies act on, from uncovering an 11.3% margin decline masked by revenue growth, to automating reporting pipelines that deliver KPI and risk alerts without manual dashboard checks. Practical experience in Advanced Excel, SQL, Power BI, and Python.

SKILLS

Programming & Analysis: SQL (MS SQL Server), Query Optimization, Python (Pandas), ETL Pipelines, Data Modeling

BI & Visualization: Excel, Power BI, DAX, Power Query, Dashboard Design

Cloud & Automation: Databricks, Docker, n8n, Git/GitHub, Vector Databases (Pinecone)

PROJECTS

[Maven Toys Sales & Inventory Analysis](#) | SQL Server, Power BI, DAX, Power Query | [Live Dashboard](#)

- Built an end-to-end sales and inventory analysis on **829K transaction records** across **50 retail stores** by modeling a star-schema in SQL Server, uncovering that gross margin fell **11.3%** despite **30.9%** revenue growth.
- Designed a **5-page interactive Power BI dashboard** with drill-through store detail pages and custom DAX measures (YoY growth, margin %, stockout risk), giving stakeholders store-level and category-level visibility.
- Identified that the top revenue category (**Toys, 35% of revenue**) carried the lowest margin (**21%**) while Electronics led on margin, translating the finding into a category-mix and inventory recommendation.

[Automated Transaction Monitoring Pipeline](#) | Databricks (PySpark/SQL), n8n, Telegram Bot API

- Engineered a cloud data pipeline in Databricks to process **6.3M synthetic mobile money transactions**, building a star-schema model to support daily and weekly KPI reporting.
- Developed **4 explainable, rule-based risk triggers** (fraud rate breach, transfer-to-cash-out chains, volume collapse, transaction spikes) validated against ground-truth fraud labels, surfacing a high-risk laundering chain worth **~\$518K**.
- Automated reporting via **3 cron-scheduled n8n workflows**: a daily KPI monitor, a weekly performance summary and a conditional alert engine that stays silent unless a risk trigger fires, delivering all reports directly to Telegram without manual dashboard checks.

[RAG Pipeline & AI Chatbot](#) | n8n, Google Gemini Embeddings, Pinecone, Google Gemini

- Designed a Retrieval-Augmented Generation chatbot workflow in n8n, connecting Google Drive as a live document source with an automated ingestion pipeline covering download, recursive character text splitting, and embedding.
- Configured an AI Agent node with a Google Gemini chat model, conversational memory, and a Pinecone retrieval tool, allowing the chatbot to answer natural language questions by querying indexed documents in real time.

EDUCATION

Metropolitan University, Sylhet | 2021 - 2025

B.Sc. in Computer Science and Engineering — CGPA: 3.96 / 4.00

CERTIFICATIONS

Data Analyst Associate — DataCamp

Data Analyst in Databricks — DataCamp

ACHIEVEMENTS

- Recipient of the Chairman's Scholarship and Vice Chancellor's Scholarship at Metropolitan University.
- Solved **100+** SQL problems across StrataScratch, DataLemur, HackerRank, and LeetCode — documented in the SQL Practice Portfolio on [GitHub](#).